

Discovering Properties
of
Real Numbers and Functions:
Rigor with Vigor

Anton Zettl

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To Harry Lucas
and
the mathematical descendants
of R. L. Moore.

Abstract

Students will discover and prove properties of real numbers and functions by presenting their own proofs and ideas in class. In this way, they will not only examine numbers and functions from fresh points of view, but also learn to use language precisely, make logical arguments, and give presentations to groups.

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Introduction

These notes are for a ‘discovery’ learning course. Problems/Theorems, based on these notes, are presented in class and solutions/proofs are, at some later date, presented on the blackboard by the students. Each student is expected to work on these Problems/Theorems on his/her own without consulting any book or any other person and will have numerous opportunities to present things on the board. Besides learning basic properties of numbers and functions, students learn to be careful and precise in their use of language, make and defend presentations to a group, prove theorems, criticize other presentations, etc.

These notes provide a solid foundation for the study of calculus. They can be used in a classroom setting and they can also be used for self study. In either case they prepare the student for understanding $\varepsilon - \delta$ definitions. Even R. L. Moore and H. S. Wall, the legendary IBL teachers at the University of Texas, Austin, did not use $\varepsilon - \delta$ definitions in their calculus classes. Also some of the most prominent IBL teachers of today do not use $\varepsilon - \delta$ definitions in their freshmen calculus classes. Nevertheless, it is this writer’s conviction that serious students in high school and college can learn to understand, and therefore use, $\varepsilon - \delta$ definitions, provided they first learn:

1. inequalities and
2. absolute values.

To do this the students must be given an appropriate amount of time to absorb this material before they can reasonably be expected to understand rigorous definitions of convergence. It is here that traditional calculus courses fail: the students are not given sufficient time to absorb the prerequisites to an understanding of the rigorous definition of a limit.

It is this writer’s belief that sequences of real numbers provide the most natural setting for the introduction of the limit concept. Once limits of sequences are well understood the ‘move up’ to continuity, differentiability, etc. is well within the student’s grasp.

Up to, but not including, the section on continuous functions, these notes were class tested in a course taught through the Harvard Extension College in the summer of 2004 and at Texas A & M University, Kingsville during the academic year 2005/2006. The later course was listed as a ‘theorem proving’ course; i.e. a course intended to teach students the art of proving theorems. Another class test occurred some time ago in an IBL (before this acronym was used) advanced calculus course the author taught at Northern Illinois University. Thus these notes can also be used as a ‘front end’ for a course in real analysis.

A reader considering using these notes to teach her/his first discovery learning course is advised to consult the Educational Advancement Foundation for supporting materials e.g.

pamphlets. Here I will limit myself to two basic tips: (i) Learn your students names the first day (or before, if possible). (ii) Be patient and positive. Typically, a discovery-learning course starts slowly and ends with a fast pace. The first week or two some students may say or do ‘foolish’ things. Be patient, courteous, and respectful and insist the same from all students. Make sure, in a polite but firm and clear manner, that the student understands where she/he went wrong. The notes are written in a rather ‘dry’ manner. Make them fit your personality and the personality of your students with suitable remarks/comments, particularly at the beginning and end of a topic.

Prerequisites: There are no specific prerequisites for this course, other than a certain level of maturity, desire to learn and willingness to work independently. This course is open to students of any age from 10 to 100. These notes were specifically (and carefully) written to be suitable for high school students and teachers, as well as for college students. In particular they are suitable for people who have had a traditional calculus course (in high school or college) and for students who have not had any calculus or pre-calculus course.

Chapter 1

REAL NUMBERS

1.1 Field Axioms

The system of real numbers $(\mathbb{R}, +, \cdot, <)$ has the following basic properties, which we call Axioms.

Axiom 1. For each a, b in $\mathbb{R}$, $a + b$ is in $\mathbb{R}$ and $a \cdot b$ is in $\mathbb{R}$ and

$$a + b = b + a, \quad a \cdot b = b \cdot a.$$

Axiom 2. For each a, b, c in $\mathbb{R}$, we have

$$(a + b) + c = a + (b + c) \text{ and } (a \cdot b) \cdot c = a \cdot (b \cdot c).$$

Axiom 3. There is a number 0 in $\mathbb{R}$ with the property that for every a in $\mathbb{R}$

$$a + 0 = a.$$

Theorem 1. If a number θ in $\mathbb{R}$ has the property that $\theta + a = a$ for every a in $\mathbb{R}$, then $\theta = 0$.

Axiom 4. For each a in $\mathbb{R}$ there is a number b in $\mathbb{R}$ such that

$$a + b = 0.$$

Theorem 2. Suppose each of a, b, c is in $\mathbb{R}$ and $a + b = 0$ and $a + c = 0$. Then $b = c$.

Definition 1. Suppose a is in $\mathbb{R}$. By Theorem 2 there is one and only one b in $\mathbb{R}$ such that $a + b = 0$. This b is denoted by $-a$.

Axiom 5. There is a number 1 in $\mathbb{R}$ such that if a is in $\mathbb{R}$, then

$$a \cdot 1 = a.$$

Theorem 3. Suppose there is a number u in $\mathbb{R}$ such that if a is in $\mathbb{R}$ and $a \neq 0$, then $a \cdot u = a$. Then $u = 1$.

Axiom 6. $1 \neq 0$.

Axiom 7. If a is in $\mathbb{R}$ and a is not 0, then there is a number b in $\mathbb{R}$ such that $a \cdot b = 1$.

Theorem 4. Suppose each of a, b, c is in $\mathbb{R}$, a is not 0, $a \cdot b = 1$ and $a \cdot c = 1$. Then $c = b$.

Definition 2. Suppose a is in $\mathbb{R}$ and a is not 0. By Theorem 4 there is one and only one b in $\mathbb{R}$ with the property that $a \cdot b = 1$; this b is denoted by $\frac{1}{a}$ or a^{-1} .

Axiom 8. Suppose each of a, b, c is in $\mathbb{R}$, then

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

1.2 Order Axioms

Axiom 9. If each of a, b is in $\mathbb{R}$, then exactly one of the following three statements is true:

$$a < b \text{ or } a = b \text{ or } b < a.$$

Axiom 10. For each a, b, c in $\mathbb{R}$, if $a < b$ and $b < c$, then $a < c$.

Axiom 11. For each a, b, c in $\mathbb{R}$, if $a < b$, then $a + c < b + c$.

Axiom 12. If each of a, b, c is in $\mathbb{R}$, $a < b$ and $0 < c$, then $a \cdot c < b \cdot c$.

1.3 Consequences of the Field and Order Axioms

A couple of additional axioms will be stated later. Other properties of the real number system we will prove as consequences of the axioms. These properties are stated as Theorems such as Theorems 1 to 4 above.

Remark 1. We write $a - b = a + (-b)$.

Theorem 5. Let a, b, c be in $\mathbb{R}$. Then

i $-(-a) = a$.

ii $a \cdot 0 = 0$.

iii $(-1) \cdot a = -a$.

iv If $a < b$, then $-b < -a$.

v $0 < 1$.

vi If $0 < a < b$, then $0 < \frac{1}{b} < \frac{1}{a}$.

vii $a^2 \geq 0$.

viii If $a \cdot b = 0$, then $a = 0$ or $b = 0$.

ix

$$2 \cdot a \cdot b \leq a^2 + b^2.$$

x If $0 < a < b$, then $0 < a^2 < b^2$.

xi If $0 < a < b$, then $0 < \sqrt{a} < \sqrt{b}$. (Here we assume that each positive number has one and only one positive square root. We will prove this later, after we have added another axiom which is called the ‘least upper bound’ or ‘completeness’ axiom.)

Let $2 = 1 + 1$, $3 = 2 + 1$, $4 = 3 + 1$, etc.

Notation 1.

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

$$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

We call $\mathbb{N}$ the positive integers, $\mathbb{N}_0$ the nonnegative integers and $\mathbb{Z}$ the integers.

1.4 Mathematical Induction

Axiom 13 (Induction). If P is a subset of $\mathbb{N}$ with the following two properties:

1. 1 is in $\mathbb{P}$,
2. If k is in $\mathbb{P}$, then $k + 1$ is in $\mathbb{P}$;

then $P = \mathbb{N}$.

Theorem 6. If n is a positive integer, then

a

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}.$$

b

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

c

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n^2}{n(n+1)}.$$

d

$$1 + 3 + 5 + 7 + \dots + (2n-1) = n^2.$$

e $n^3 + 5n$ is divisible by 6.

f If n is a positive integer greater than 5, then

$$n^2 < 2^n.$$

g If a is in $\mathbb{R}$ and $a \neq 1$, then

$$1 + a + a^2 + a^3 + \cdots + a^n = \frac{1 - a^{n+1}}{1 - a}.$$

h If a is in $\mathbb{R}$ and b is in $\mathbb{R}$ and n is a positive integer, then

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b^1 + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}a^1b^{n-1} + b^n$$

where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

and $0! = 1$, $1! = 1$, $2! = 2$, $3! = 3 \cdot 2$, $4! = 4 \cdot 3 \cdot 2$, $n! = n(n-1)!$, $n = 5, 6, 7, \dots$

1.5 Absolute Values

Definition 3 (Absolute Value). If a is in $\mathbb{R}$, then

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Theorem 7. Suppose each of a, b is in $\mathbb{R}$. Then

a

$$|a + b| \leq |a| + |b|.$$

b

$$|a \cdot b| = |a| \cdot |b|.$$

c

$$|a - b| \geq ||a| - |b||$$

d If $\varepsilon > 0$, then $|a| < \varepsilon$ if and only if $-\varepsilon < a < \varepsilon$.

e Assume $\varepsilon > 0$. Then the following four statements are equivalent:

1. $|a - b| < \varepsilon$,
2. $-\varepsilon < a - b < \varepsilon$,
3. $a - \varepsilon < b < a + \varepsilon$
4. $b - \varepsilon < a < b + \varepsilon$.

1.6 Complex Numbers

There are other number systems besides $\mathbb{R}$, perhaps the most important of these is the complex number system. Here we give a very brief introduction to the complex number system.

Definition 4. The system of complex numbers $(\mathbb{C}, \oplus, \otimes)$ is defined as follows: $\mathbb{C}$ is the set of all pairs of real numbers, if each of (a, b) and (c, d) is in $\mathbb{C}$, then

$$\begin{aligned}(a, b) \oplus (c, d) &= (a + b, c + d), \\ (a, b) \otimes (c, d) &= (ac - bd, bc + ad).\end{aligned}$$

Remark 2. Note that the system of complex numbers, in contrast to the real numbers, does not include an analogue of the order relation ' $<$ '.

Problem 1. Show that the system of complex numbers satisfies the field axioms including Theorems 1 to 4. Which other theorems, established above, have a natural analogue for the system of complex numbers?

Problem 2. Define

$$|(a, b)| = \sqrt{a^2 + b^2}$$

and show that parts (a), (b) (c) of Theorem 7 hold. (Note that the absolute value of a complex number is a real number.)

Definition 5. Let

$$i = (0, 1).$$

Problem 3. Show that $i^2 = (-1, 0)$ and any $z = (a, b)$ in $\mathbb{C}$ has the representation

$$z = (a, 0) \oplus i \otimes (b, 0).$$

How do the complex numbers of the form $(a, 0)$ compare with the real numbers?

Chapter 2

Completeness Property

Definition 6. a A subset of $\mathbb{R}$ is bounded above if there is a number which is greater than or equal to every number in the set. In other words, if S is a set of numbers we say that S is bounded above if there is a number u with the property that if x is in S , then $x \leq u$. Such a number u is called an upper bound of S .

b A subset of $\mathbb{R}$ is bounded below if there is a number which is less than or equal to every number in the set. In other words, if S is a set of numbers we say that S is bounded below if there is a number l with the property that if x is in S , then $l \leq x$. Such a number l is called a lower bound of S .

c Suppose S is a set which is bounded above. A number b is called a least upper bound of S if it has the following two properties: Suppose S is a set of real numbers which is bounded above. A number b is called a least upper bound of S if it has the following two properties:

1. it is an upper bound,
2. if u is an upper bound of S , then $b \leq u$.

d Suppose S is a set which is bounded below. A number g is called a greatest lower bound of S if it has the following two properties:

1. it is a lower bound of S ,
2. if l is a lower bound of S , then $l \leq g$.

Definition 7. A subset of $\mathbb{R}$ is bounded if it has an upper bound and a lower bound.

Axiom 14 (Least Upper Bound Axiom). Every subset of $\mathbb{R}$ which is bounded above has a least upper bound.

Theorem 8. 1. Suppose S is a set of numbers which is bounded above. Then it has one and only one least upper bound.

2. Suppose S is a set of numbers which is bounded below. Then it has one and only one greatest lower bound.

Theorem 9. Suppose S is a set of real numbers which is bounded above and u is its least upper bound. If ε is a positive number, then there is a number b in S such that

$$u - \varepsilon < b \leq u.$$

Theorem 10. Suppose S is a set of real numbers which is bounded below and g is its greatest lower bound. If ε is a positive number, then there is a number b in S such that

$$g \leq b < g + \varepsilon.$$

Theorem 11. a The set $\mathbb{N} = \{1, 2, 3, \dots\}$ is not bounded above.

b The set $S = \{-1, -2, -3, \dots\}$ is not bounded below.

Theorem 12. For each a in $\mathbb{R}$, there is an integer n such that

$$n \leq a < n + 1.$$

Theorem 13. If $\varepsilon > 0$, there exists a positive integer n such that

$$0 < \frac{1}{n} < \varepsilon.$$

Theorem 14. Suppose $a > 0$. There is a number $t > 0$ such that $t^2 \leq a$.

Theorem 15. Suppose $a > 0$. There is one and only one $r > 0$ such that $r^2 = a$. This number r is called the square root of a and is denoted by $\sqrt{a}$.

Definition 8. A real number b is said to be rational if there exists an integer n and an integer m with $m \neq 0$, such that

$$b = \frac{n}{m}.$$

A real number is said to be irrational if it is not rational.

Theorem 16. The $\sqrt{2}$ is not a rational number.

Problem 4. Show that the rational numbers satisfy the field axioms, Theorems 1 to 4 and the order axioms. Which other theorems are true when the set of real numbers $\mathbb{R}$ is replaced by the set of rational numbers? Do the rational numbers satisfy the completeness axiom?

Chapter 3

Sequences

Definition 9. A function is a set of ordered pairs of objects with the property that no two pairs have the same first term. The objects can be anything, but, unless specifically stated otherwise, we will take them to be real numbers. The set of all first terms is called the domain of the function, the set of all second terms the range. A function whose domain is the set of positive integers $\mathbb{N} = \{1, 2, 3, \dots\}$ is called a sequence.

Notation 2. Instead of writing $\{(n, a_n) : n \in \mathbb{N}\} = \{(1, a_1), (2, a_2), (3, a_3), \dots\}$ for a sequence we often, but not always, just write $\{a_n : n \in \mathbb{N}\}$ or $a_1, a_2, a_3, \dots$.

To gain some familiarity with sequences we list a few examples.

Example 1. 1. $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$

2. $1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots$

3. $1, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \dots$

4. $1, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{4}}, \dots$

5. $\{a_n = -5 + \frac{3}{n} : n \in \mathbb{N}\}$

6. $\{a_n = 7 + (-1)^n \frac{1}{n} : n \in \mathbb{N}\}$

7. $1, -1, 1, -1, \dots$

8. $0, 1, 0, 1, 0, 1, \dots$

9. $0, 1, 0, \frac{1}{2}, 0, \frac{1}{3}, 0, \frac{1}{4}, \dots$

10. $\{a_n : n \in \mathbb{N}, \text{ where } a_n \text{ is the } n\text{-th prime}\}$; the first few terms are: 2, 3, 5, 7, 11, 13, 17, 19, 23, ..., determining the 97-th term takes some work. There is no explicit formula for the n -th term which is valid for all n . (Problem : Prove that there are an infinite number of primes, thus showing that this is indeed a sequence.)

11. $1, 1 + \frac{1}{2}, 1 + \frac{1}{2} + \frac{1}{3}, 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}, 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}, \dots$

12. $\{a_n = (\sqrt{n+1} - \sqrt{n}) : n \in \mathbb{N}\}$

13. $\{a_n = (1 + \frac{1}{n})^n : n \in \mathbb{N}\}$
14. Assume $0 < b < 1$. Consider the sequence $\{b_n = b^n : n \in \mathbb{N}\}$
15. Assume $1 < b$. Consider the sequence $\{b_n = b^n : n \in \mathbb{N}\}$
16. Assume $0 < b < 1$. Consider the sequence $\{b_n = \sqrt[n]{b} : n \in \mathbb{N}\}$
17. We define a sequence $\{a_n : n \in \mathbb{N}\}$ recursively as follows: $a_1 = 1$, $a_2 = \sqrt{2}$, for each $n \in \mathbb{N}$, $n > 2$, $a_n = \sqrt{2a_{n-1}}$,
18. $\{a_n = \frac{2n-9}{3n-28} : n \in \mathbb{N}\}$
19. $\{a_n = \frac{4n+7}{5n+8} : n \in \mathbb{N}\}$
20. $\{a_n = \frac{5n+1}{10n^2+6} : n \in \mathbb{N}\}$
21. $\{a_n = \frac{3n^2-27}{2n^2+9} : n \in \mathbb{N}\}$
22. $\{a_n = \frac{-5n^2+7n-97}{3n^2+8n-16} : n \in \mathbb{N}\}$
23. $\{a_n = \frac{3n^3+1}{7n^2+9} : n \in \mathbb{N}\}$
24. $\{a_n = \frac{\sqrt{n}+1}{n+2} : n \in \mathbb{N}\}$
25. $\{a_n = \frac{n+3}{\sqrt{n+5}} : n \in \mathbb{N}\}$
26. $\{a_n : n \in \mathbb{N}\}$
27. $\{a_n : n \in \mathbb{N}\}$
28. $\{a_n = \sqrt[n]{n} : n \in \mathbb{N}\}$
29. $\{a_n = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots + \frac{1}{n^2}, n = 1, 2, 3, \dots\}$

Below we will refer to these sequences by their number on this list.

It is convenient to classify sequences into types.

Definition 10 (Types of Sequences). A sequence $\{a_n : n \in \mathbb{N}\}$ is said to be

1. *nondecreasing* if $a_n \leq a_{n+1}$ for each $n \in \mathbb{N}$,
2. *increasing* if $a_n < a_{n+1}$ for each $n \in \mathbb{N}$,
3. *nonincreasing* if $a_n \geq a_{n+1}$ for each $n \in \mathbb{N}$,
4. *decreasing* if $a_n > a_{n+1}$ for each $n \in \mathbb{N}$,
5. *bounded above* if its range is bounded above,
6. *bounded below* if its range is bounded below,
7. *bounded* if its range is bounded.

Problem 5. Find a sequence of each type from 1 to 7.

Theorem 17. Suppose that a number $a \in \mathbb{R}$ has the following property: for every positive number ε it is true that $|a| < \varepsilon$. Then $a = 0$.

Definition 11. Given a sequence $\{a_n : n \in \mathbb{N}\}$ and a number $a \in \mathbb{R}$, we say that the sequence $\{a_n : n \in \mathbb{N}\}$ converges to a if for any positive number ε there is a positive integer M such that if n is any integer greater than M , then

$$|a_n - a| < \varepsilon.$$

We say that the sequence $\{a_n : n \in \mathbb{N}\}$ converges or is convergent if there is a number a such that the sequence $\{a_n : n \in \mathbb{N}\}$ converges to a .

Problem 6. Given a sequence $\{a_n : n \in \mathbb{N}\}$ and a number $a \in \mathbb{R}$, what does it mean to say that the sequence $\{a_n : n \in \mathbb{N}\}$ does not converge to a ? What does it mean to say that the sequence $\{a_n : n \in \mathbb{N}\}$ does not converge?

Theorem 18. Let $\{a_n : n \in \mathbb{N}\}$ be a sequence and let $a \in \mathbb{R}$ and $b \in \mathbb{R}$. If the sequence $\{a_n : n \in \mathbb{N}\}$ converges to a and the sequence $\{a_n : n \in \mathbb{N}\}$ converges to b , then $a = b$.

Problem 7. Prove that the sequence $\{a_n = \frac{5}{n} : n \in \mathbb{N}\}$ converges to 0.

Theorem 19. Let $c \in \mathbb{R}$. If the sequence $\{a_n : n \in \mathbb{N}\}$ converges to a , then the sequence $\{b_n = ca_n : n \in \mathbb{N}\}$ converges to $c \cdot a$.

Theorem 20. Every nondecreasing sequence which is bounded above converges.

Theorem 21. Every nonincreasing sequence which is bounded below converges.

Problem 8. Find a sequence which converges but is neither nondecreasing nor nonincreasing.

Theorem 22. Suppose

1. The sequence $\{a_n : n \in \mathbb{N}\}$ converges to $a \in \mathbb{R}$,
2. The sequence $\{b_n : n \in \mathbb{N}\}$ converges to $a \in \mathbb{R}$,
3. For each $n \in \mathbb{N}$, $a_n \leq c_n \leq b_n$;

Then the sequence $\{c_n : n \in \mathbb{N}\}$ converges to $a \in \mathbb{R}$.

Theorem 23. Every convergent sequence is bounded.

Theorem 24. Suppose the sequence $\{a_n : n \in \mathbb{N}\}$ converges to $a \in \mathbb{R}$, and $a_n \geq 0$ for each $n \in \mathbb{N}$. Then $a \geq 0$ and $\{\sqrt{a_n} : n \in \mathbb{N}\}$ converges to $\sqrt{a}$.

Theorem 25. Suppose each of a, b is in $\mathbb{R}$, and each of $\{a_n : n \in \mathbb{N}\}$, $\{b_n : n \in \mathbb{N}\}$ is a sequence. Assume that $\{a_n : n \in \mathbb{N}\}$ converges to a and $\{b_n : n \in \mathbb{N}\}$ converges to b . Then

1. the sequence $\{c_n = a_n + b_n : n \in \mathbb{N}\}$ converges to $a + b$,
2. the sequence $\{c_n = a_n \cdot b_n : n \in \mathbb{N}\}$ converges to $a \cdot b$,

3. if $b_n \neq 0$ for each $n \in \mathbb{N}$ and $b \neq 0$, then the sequence $\{\frac{a_n}{b_n} : n \in \mathbb{N}\}$ converges to $\frac{a}{b}$.

An important notion for sequences is the concept of a subsequence.

Definition 12. Given a sequence $\{a_n : n \in \mathbb{N}\}$ we say that the sequence $\{b_n : n \in \mathbb{N}\}$ is a subsequence of $\{a_n : n \in \mathbb{N}\}$ if there exists an increasing sequence of positive integers $n_1, n_2, n_3, \dots$ such that $b_1 = a_{n_1}, b_2 = a_{n_2}, b_3 = a_{n_3}, \dots$

Theorem 26. If $\{a_n : n \in \mathbb{N}\}$ converges to a number a , then every subsequence of $\{a_n : n \in \mathbb{N}\}$ converges to a .

Notation 3 (Intervals). For a, b in $\mathbb{R}$ with $a < b$, let

$$\begin{aligned} [a, b] &= \{t \in \mathbb{R} : a \leq t \leq b\}, \\ (a, b) &= \{t \in \mathbb{R} : a < t < b\}, \\ [a, b) &= \{t \in \mathbb{R} : a \leq t < b\}, \\ (a, b] &= \{t \in \mathbb{R} : a < t \leq b\}, \\ [a, \infty) &= \{t \in \mathbb{R} : a \leq t\}, \\ (-\infty, b] &= \{t \in \mathbb{R} : t \leq b\}, \\ (-\infty, \infty) &= \mathbb{R} = \{t : -\infty < t < \infty\}. \end{aligned}$$

Theorem 27. Let A, B be in $\mathbb{R}$ and assume that $A < B$. Let $C = (A + B)/2$. Let $[a_1, b_1]$ be one of the two subintervals $[A, C]$ or $[C, B]$. Repeat this process starting with $[a_1, b_1]$ to construct $[a_2, b_2]$. Now starting with $[a_2, b_2]$ construct $[a_3, b_3]$ in the same way and continue this process to obtain a sequence of intervals $\{[a_n, b_n] : n \in \mathbb{N}\}$. Then there is one and only one number, say h which is contained in every one of the intervals $\{[a_n, b_n] : n \in \mathbb{N}\}$.

Theorem 28. Every bounded sequence has a convergent subsequence.

Definition 13. A sequence $\{a_n : n \in \mathbb{N}\}$ is Cauchy, if for any positive number ε there is a positive integer M such that if n and m are positive integers greater than M then

$$|a_n - a_m| < \varepsilon.$$

Theorem 29. Every convergent sequence is Cauchy.

Theorem 30. Every Cauchy sequence is bounded.

Theorem 31. Every Cauchy sequence has a convergent subsequence.

Theorem 32. Every Cauchy sequence converges.

3.1 Continuous Functions

Recall that a function is a set of pairs of objects with the property that no two pairs have the same first term. In general these ‘objects’ can be anything but in this section, unless explicitly stated otherwise, the first and second term of each pair is a real number. (The set of all pairs whose first term is an occupied chair in this room (at this time) and whose second terms consist of the occupants of these chairs is a function provided no chair is

occupied by two people.) The set of all first terms of a function f is called the domain of f , denoted by D_f , the set of all second pairs is called the range of f and is denoted by R_f .

A sequence is a special kind of function whose domain is the positive integers. In this section we encounter functions with other domains. Some of these domains contain an infinite number of elements but cannot be enumerated with the positive integers. So it is helpful to introduce some additional notation. Let

$$f = \{(t, t^2) : t \in \mathbb{R}\}$$

Then f is a function with domain $D_f = \mathbb{R}$. What is the range of f ? This function can also be described by writing $f(t) = t^2$ for $t \in \mathbb{R}$. This tells us that the first term can be any number and the second term is always the square of the first term. In general, given any function f , $f(t)$ denote the second term of that pair in f whose first term is t . We may want to consider another function, say g , defined by the same formula but only for a restricted set of first terms: $g(t) = t^2$ for $2 \leq t < 7$. This function has domain $[2, 7)$ and is therefore different from the function f . What is the range of g ? There is a general convention in Mathematics that when a function is specified by a formula and the domain is not explicitly given it is understood that the domain consists of all elements for which the formula makes sense.

A function whose range consists of only one number is called a constant function.

We list a few examples of functions; many more will be encountered below.

Example 2.

1. $f(t) = t$,
2. $f(t) = 3t - 8$,
3. $f(t) = t^2$,
4. $f(t) = t^2 + 5$,
5. $f(t) = \sqrt{t}, t \geq 0$,
6. $f(t) = \frac{1}{t}$,
7. $f(t) = \frac{2}{t^2 - 3}$,
8. $f(t) = \frac{2}{t^2 - 9}$,
9. $f(t) = \frac{t+3}{t^2 - 9}$,
10. $f(t) = \frac{2}{t^2 + 3}$,
11. $f(t) = t^3 + 1$,
12. $f(t) = t^3 - 5t^2 + 7t - 9$,
13. $f(t) = a_n t^n + a_{n-1} t^{n-1} + a_{n-2} t^{n-2} + \dots + a_1 t + a_0$. A function of this type with $a_j \in \mathbb{R}$, n a positive integer, and $a_n \neq 0$ is called a polynomial of degree n . (A constant function can be thought of as a polynomial of degree 0.)

14. $f(t) = |t|,$

15. $f(t) = \frac{1}{|t|},$

16. $f(t) = [t],$ where $[t]$ denotes the greatest integer $\leq t$.

Many new functions can be constructed from a given function or from two functions. If f is a function and k is any number then $g(t) = k \cdot f(t)$ defines another function with the same domain as f . Suppose each of f and g is a function and there is at least one number in the domain of both f and g , then $h(t) = f(t) + g(t)$ for all t in the domain of both f and g defines a function h . Similarly a product function can be defined by the formula $k(t) = f(t) \cdot g(t)$. If there is a number x in the domain of g such that $g(x) \neq 0$ then the quotient function $\frac{f}{g}$ can be defined similarly.

Definition 14. Suppose S is a subset of $\mathbb{R}$ and $p \in \mathbb{R}$. To say that p is an accumulation point of S means that every open interval which contains p also contains a point of S different from p . Note that p may or may not be in S .

Definition 15. Suppose f is a function with domain $D = D_f$ and p is an accumulation point of D . To say that f has a limit at p means that there is a number L such that if $\{t_n : n \in \mathbb{N}\}$ is any sequence converging to p and t_n is in D for each $n \in \mathbb{N}$, then the sequence $\{f(t_n) : n \in \mathbb{N}\}$ converges to L . If such a number L exists it is unique, is called the limit of f at p , and is denoted by $\lim_{t \rightarrow p} f(t) = L$.

Definition 16. Suppose f is a function with domain $D = D_f$, p is an accumulation point of D and p is in D . To say that f is continuous at p means that f has a limit at p and this limit is $f(p)$.

Problem 9. Prove that examples 1-4 are continuous at p for every number p .

Problem 10. What is the domain of example 9? At what points does example 9 have a limit? At what points is it continuous? At what point or points does it not have a limit?

Problem 11. Note that the domain of example 16 is $\mathbb{R}$. At what points does example 16 have a limit? at what points is it continuous? at what points is it not continuous?

Theorem 33. Suppose that each of f and g is a function, $D_f = D_g$, and p is an accumulation point of D_f . Suppose f has limit L at p and g has limit K at p . Then

1. $f + g$ has limit $L + K$ at p ,
2. for any r in $\mathbb{R}$ $r \cdot f$ has limit rL at p ,
3. $f \cdot g$ has limit $L \cdot K$ at p ,
4. if $g(t) \neq 0$ for each t in D_g and $K \neq 0$, then $\frac{f}{g}$ has limit $\frac{L}{K}$ at p .

Theorem 34. Suppose that each of f and g is a function, $D_f = D_g$, p is an accumulation point of D_f and p is in D_f . Suppose each of f and g is continuous at p . Then

1. $f + g$ is continuous at p ,
2. $f \cdot g$ is continuous at p ,

3. for any r in $\mathbb{R}$, $r \cdot f$ is continuous at p ,
4. if $g(t) \neq 0$ for each t in D_g , then $\frac{f}{g}$ is continuous at p .

Definition 17. Suppose f is a function and S is an interval, open, closed or half-open, which is contained in the domain of f . To say that f is continuous on S means that f is continuous at each point of S .

Theorem 35. Every polynomial function is continuous on $\mathbb{R}$.

Theorem 36. Suppose f is continuous on an interval $[a, b] \subset D_f$. Then

1. f is bounded on $[a, b]$. (This means that the set $\{f(t) : t \in [a, b]\}$ is bounded.)
2. If $f(p) > 0$ for some $p \in [a, b]$, then there is an open interval J containing p such that $f(t) > 0$ for all t in $J \cap [a, b]$.
3. If $f(p) < 0$ for some $p \in [a, b]$, then there is an open interval J containing p such that $f(t) < 0$ for all t in $J \cap [a, b]$.
4. If there are points c, d in $[a, b]$. $c < d$ such that $f(c) < 0$ and $f(d) > 0$ then there is a point h in (c, d) such that $f(h) = 0$.
5. If there are points c, d in $[a, b]$. such that $f(c) < f(d)$ and $y \in (f(c), f(d))$ then there is a point h in (c, d) such that $f(h) = y$.
6. There is a point c in $[a, b]$. such that $f(c) \leq f(t)$ for every t in $[a, b]$. A continuous function on a compact interval assumes its minimum value on that interval.
7. There is a point d in $[a, b]$. such that $f(t) \leq f(d)$ for every t in $[a, b]$. A continuous function on a compact interval assumes its maximum value on that interval.

Problem 12. Show that there is a number r such that $r^3 - 7r^2 + 35r + 48 = 0$.

Theorem 37. Every polynomial of odd degree has at least one real root. In other words: If f is a polynomial of degree n and n is odd, then there is a real number r such that $f(r) = 0$.

3.2 Derivatives

Definition 18. Suppose f is a function with domain $D = D_f$, p is an accumulation point of D and p is in D . To say that f is differentiable at p means that given any sequence $\{t_n : n \in \mathbb{N}\}$ satisfying the following three conditions:

1. $t_n \in D_f$, $n \in \mathbb{N}$,
2. $t_n \neq p$, $n \in \mathbb{N}$,
3. $\{t_n : n \in \mathbb{N}\}$ converges to p ,

it follows that the sequence

$$\frac{f(t_n) - f(p)}{t_n - p}$$

converges. The limit L of this sequence is called the derivative of f at p and is denoted by $f'(p)$, i.e. $f'(p) = L$.

Problem 13. For $f(t) = t^2$, show that $f'(3) = 6$, $f'(5) = 10$, $f'(-4) = -8$. Formulate and prove a general result about $f'(p)$.

Problem 14. Find a function f and a point p such that f is not differentiable at p .

Theorem 38. Suppose that each of f and g is a function with domain $D = D_f = D_g$, p is an accumulation point of D and p is in D . Assume that each of f and g is differentiable at p . Then

1. $f + g$ is differentiable at p and

$$(f + g)'(p) = f'(p) + g'(p),$$

2. $f \cdot g$ is differentiable at p , and

$$(f \cdot g)'(p) = f'(p) \cdot g(p) + f(p) \cdot g'(p),$$

3. for any $k \in \mathbb{R}$, $k \cdot f$ is differentiable at p , and

$$(k \cdot f)'(p) = k \cdot f'(p),$$

4. if $g(t) \neq 0$ for all t in D_g , then $\frac{f}{g}$ is differentiable at p and

$$\left(\frac{f}{g}\right)'(p) = \frac{f(p) \cdot g'(p) - f'(p) \cdot g(p)}{(g(p))^2}.$$

Problem 15. Show that every polynomial is differentiable at every point and find a formula for its derivative.

Theorem 39. Show that the functions $f(t) = \frac{1}{t}$ and $g(t) = \sqrt{t}$ are differentiable at each point of their domain and find their derivatives.

Theorem 40. If f is differentiable at p then f is continuous at p . The converse statement is false.

Definition 19. Assume that $[a, b] \in D_f$. To say that f has a relative maximum at a point p in (a, b) means that there is an open interval J containing p such that $f(t) \leq f(p)$ for every t in J . To say that f has a relative minimum at a point p in (a, b) means that there is an open interval J containing p such that $f(t) \geq f(p)$ for every t in J .

Theorem 41. Suppose f is a function whose domain includes the interval $[a, b]$.

1. Assume that f is differentiable at a point p in (a, b) and $f'(p) > 0$. Then there is an open interval J containing p such that if h, k are in J and $h < p < k$ the $f(h) < f(p) < f(k)$.
2. Assume that f is differentiable at a point p in (a, b) and $f'(p) < 0$. Then there is an open interval J containing p such that if h, k are in J and $h < p < k$ the $f(h) > f(p) > f(k)$.
3. If f has a relative maximum at a point p in (a, b) and f is differentiable at p then $f'(p) = 0$.

4. If f has a relative minimum at a point p in (a,b) and f is differentiable at p then $f'(p) = 0$.
5. Assume that f is continuous on $[a,b]$ and differentiable at each point of the open interval (a,b) and there are points c,d in (a,b) such that $f(c) = 0$ and $f(d) = 0$, then there is a point h between c and d such that $f'(h) = 0$.
6. Assume that f is continuous on $[a,b]$ and differentiable at each point of the open interval (a,b) , then there is a point h between c and d such that

$$f'(h) = \frac{f(b) - f(a)}{b - a}.$$

Next we mention a few problems which serve to clarify and illuminate the above results.

Problem 16. 1. Show that the cubic equation

$$x^3 + x + 5 = 0$$

has one and only one real root.

2. Show that for each $h > 0$,

$$\sqrt{1+h} < 1 + \frac{h}{2}.$$

Remark 3. The derivative section is just a brief introduction. We plan to expand these notes to include the basic facts of calculus and publish the expanded version either in the form of another article or, possibly, in book form.